

30 Heisenberg's Uncertainty Principle

$$\Delta x \Delta p \geq h$$

$$\begin{aligned}\Delta p &= (1.00\%)(6.4 \times 10^{-23}) \\ &= 6.4 \times 10^{-25} \text{ N s}\end{aligned}$$

$$\begin{aligned}\Delta x &= \frac{h}{\Delta p} = \frac{6.63 \times 10^{-34}}{6.4 \times 10^{-25}} \\ &= 1.03 \times 10^{-9} \text{ m}\end{aligned}$$